

DANIEL GILL

Computational Health Physics · Research Reactor Operations · Radiation Protection
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SUMMARY

Research reactor operator and health physics graduate researcher specializing in computational radiation transport, dose assessment, and instrumentation. Combines hands-on reactor operations with Monte Carlo modeling (MCNP6.3, OpenMC), Python-based data acquisition and analysis, and shielding computation to bridge experimental reactor work and computational health physics. Experienced in neutron dosimetry, beam-port characterization, radiation safety and regulatory compliance, and computing for transport simulations.

TECHNICAL SKILLS

Computational & Modeling: MCNP6.3, OpenMC, neutron & photon transport, radiation shielding analysis (buildup factors, dose conversion factors), spectral & damage-function analysis, and parallel computing (Slurm, OpenMPI)

Programming & Tools: Python, Java, C++, JavaScript, HTML; Git, Docker, Linux

Health Physics & Radiation Protection: Dose assessment, neutron dosimetry, electronics irradiation & displacement-damage characterization, radiation waste management, radiation safety & regulatory compliance

Reactor Operations: Control-room operations, reactor instrumentation & equipment monitoring, surveillance & log keeping, hazardous materials handling

PROFESSIONAL EXPERIENCE

University of Massachusetts Lowell Research Reactor

Lowell, MA

Reactor Operator (promoted from Auxiliary Operator)

April 2024 – Present

- Operate the 1 MW UMLRR research reactor for experimental irradiations and external client services, maintaining strict compliance with technical specifications and radiation protection procedures.
- Perform daily reactor startup checks, instrumentation and safety-channel verifications, and equipment surveillances; document results in operational logs, forms, and reports.
- Developed an MCNP6.3 thermal-column moderator design study (M.S. thesis) to engineer a high-fluence, high-purity thermal neutron beam at the reactor's 6-inch beam port, building an end-to-end Python pipeline that generates parameterized input decks, executes batched runs, and post-processes spectral and displacement-damage results.
- Designed and built a transistor-based neutron dosimetry system pairing automated Python instrumentation control (Keithley source-measure units over GPIB) with ASTM E1855 analysis to quantify fast-neutron displacement-damage fluence.
- Developed a digital radiation waste tracking system with isotope-decay-based safe-release-date estimation, replacing a manual paper-based process and improving audit readiness.
- Maintain controlled and contamination-controlled areas to inspection readiness, contributing to consistently high audit and compliance scores.

Cinemark

Salem, NH

Lead Line Cook

February 2022 – September 2024

- Led high-volume kitchen operations and opening/closing setup while sustaining near-perfect customer satisfaction score.

RIOS Medical

Wells, ME

Intern

April 2021 – July 2022

- Prepared project presentations and reports and maintained database systems to track and analyze operational data.

RESEARCH & TECHNICAL PROJECTS

Thermal Neutron Beam Optimization via MCNP6.3 Monte Carlo Modeling

Designing a graphite thermal-column geometry with an upstream bismuth gamma filter to deliver a high-fluence thermal neutron beam (thermal-to-fast ratio $\geq 100:1$) at the UMLRR 6-inch beam port. Built a complete Python automation pipeline that generates MCNP6.3 input decks from parameterized geometries, executes batched

simulations, parses MCTAL tally output, and folds results against the ASTM E722 silicon displacement-damage function. Derived a synthetic SDEF neutron source from multigroup beam-port flux data to drive the model.

Verification of Simulated Facility Dose Profiles - UMLRR 5 MW Upgrade

Capstone

Developed a combined photon and neutron shielding calculator to verify dose profiles for the reactor's proposed 1 MW → 5 MW power upgrade. Implemented energy-dependent dose conversion factors, Berger buildup parameters for concrete, and linear attenuation coefficients to compute deep-dose-equivalent rates through facility shielding, with power-scaling to model the upgraded configuration and compare results against existing shield requirements for regulatory compliance.

Transistor-Based Neutron Dosimetry System

M.S. Thesis — In Progress

Built a complete data-acquisition and analysis system measuring fast-neutron displacement-damage fluence with 2N2222A silicon bipolar transistors per ASTM E1855. Developed Python instrumentation software interfacing a Keithley source-measure unit to automate pre-irradiation screening and post-irradiation gain measurements with iterative base-current convergence, plus a standalone fluence calculator implementing the Messenger-Spratt equation with full uncertainty propagation. Characterized TO-18 metal-can and 2N2222AUB surface-mount packages across 1×10^{10} to 6×10^{14} n/cm².

EDUCATION

University of Massachusetts Lowell

Lowell, MA

M.S., Radiological Sciences and Protection (Health / Nuclear Physics)

Expected May 2027

- Thesis: high-fluence thermal neutron beam design for the UMLRR 6-inch beam port (MCNP6.3).

University of Massachusetts Lowell

Lowell, MA

B.S., Physics

May 2026

- Dean's List (all semesters); Daniel C. Cole Endowed Physics Scholarship; Dean's Scholarship.
- President, UML Society of Physics Students; Treasurer, Chess Club; member, American Nuclear Society.

LICENSES & CERTIFICATIONS

- Reactor Operator License — U.S. Nuclear Regulatory Commission (UMass Lowell Research Reactor).
- Nuclear Criticality Safety Certification, Levels I & II — August 2025.
- Radiation Safety Training — UMass Lowell & Georgia Tech, April 2024 & August 2025.
- Forklift Certification — June 2023.